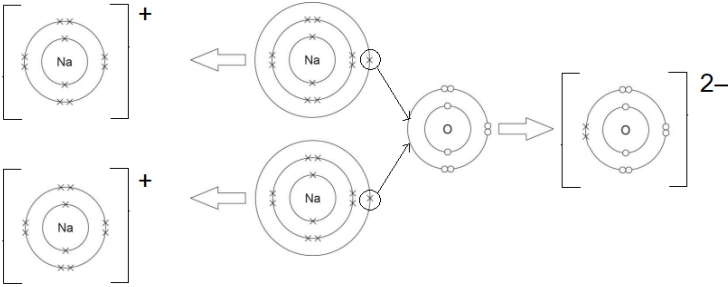
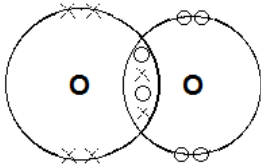


HIGHER TIER ONLY QUESTIONS

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 2        | (a) | (i)  |  |  <p>outer shell electron of both sodium atoms shown going to the outer shell of the oxygen atom (1)</p> <p>correct charges on <b>both</b> sodium ions <u>and</u> the oxide ion (1)</p>                                                        |                 | 2   |     | 2     |       |      |
|          |     | (ii) |  |  <p>award (2) for correct diagram with <b>two</b> shared pairs of electrons in the overlap <u>and</u> correct octet for both atoms</p> <p>if incorrect award (1) for overlapping oxygen atoms with <b>two</b> shared pairs in the overlap</p> |                 | 2   |     | 2     |       |      |

| Question |     |      |    | Marking details                                                                    | Marks available |     |     |       |       |      |
|----------|-----|------|----|------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) | (i)  | I  | graphite – (because it has) free/mobile/delocalised electrons (between the layers) | 1               |     |     | 1     |       |      |
|          |     |      | II | graphite – (because) the layers can slide over each other                          | 1               |     |     | 1     |       |      |
|          |     | (ii) |    | nanotubes / graphene / fullerene / buckminsterfullerene<br>accept 'bucky ball'     | 1               |     |     | 1     |       |      |
|          |     |      |    | Question 2 total                                                                   | 3               | 4   | 0   | 7     | 0     | 0    |